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Turbine support case with axial spokes and heat shields

Abstract

An aircraft engine, has: a turbine including a turbine rotor rotatable about a central axis; a scroll case having an inlet connected to a source of combustion gases and an outlet fluidly connected to the turbine, and a conduit extending around the central axis from the inlet to the outlet; a bearing housing; a turbine support case secured to the bearing housing, the scroll case disposed axially between the turbine support case and the bearing housing, the turbine support case having spokes extending through the scroll case and radially supported by the bearing housing; and heat shields distributed around the central axis in circumferential alignment with the spokes and extending axially through the scroll case for thermally shielding the spokes, a heat shield of the heat shields disposed around a spoke of the spokes, the heat shield defining an air gap surrounding the spoke inside the scroll case.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2421833	12/1946	Cronstedt et al.	N/A	N/A
3738105	12/1972	Buchelt	N/A	N/A
5105625	12/1991	Bell, III et al.	N/A	N/A
5156534	12/1991	Burgy et al.	N/A	N/A
6968697	12/2004	Nguyen et al.	N/A	N/A
7093448	12/2005	Nguyen et al.	N/A	N/A
7731426	12/2009	Meacham et al.	N/A	N/A
7931437	12/2010	Johnson	N/A	N/A
8069648	12/2010	Snyder	60/761	F01D 25/30
8272203	12/2011	Derenes et al.	N/A	N/A
9488071	12/2015	Anglin	N/A	F01D 25/164
9644494	12/2016	Hashimoto	N/A	F01D 25/12
9970295	12/2017	Fontaine et al.	N/A	N/A
9982561	12/2017	Budnick et al.	N/A	N/A
10677086	12/2019	Kobayashi et al.	N/A	N/A
10815832	12/2019	Staffier et al.	N/A	N/A
11187149	12/2020	Kobielski et al.	N/A	N/A
11643218	12/2022	Blagojevic	N/A	N/A
12065950	12/2023	Lefebvre et al.	N/A	N/A
2006/0120854	12/2005	Wakeman	415/104	F01D 25/28
2010/0061846	12/2009	Herbst	29/888.021	F01D 5/005
2011/0008162	12/2010	Yokoyama et al.	N/A	N/A
2011/0311355	12/2010	Benkler et al.	N/A	N/A
2013/0064655	12/2012	Smatloch et al.	N/A	N/A
2013/0156567	12/2012	Nagae et al.	N/A	N/A
2017/0204743	12/2016	Yokoyama et al.	N/A	N/A
2017/0218760	12/2016	Fontaine et al.	N/A	N/A
2018/0216494	12/2017	Azuma et al.	N/A	N/A
2019/0316516	12/2018	Arai	N/A	N/A
2020/0182257	12/2019	Uemura et al.	N/A	N/A
2020/0200073	12/2019	Uemura et al.	N/A	N/A
2021/0231021	12/2020	Maurya	N/A	F01D 25/24

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102014012123	12/2015	DE	N/A
2002349276	12/2001	JP	N/A
3757999	12/2005	JP	N/A
4370232	12/2008	JP	N/A
2016184549	12/2015	WO	N/A
2016184550	12/2015	WO	N/A

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Background/Summary

TECHNICAL FIELD

(1) The disclosure relates generally to aircraft engines and, more particularly, to a turbine support case for such engines.

BACKGROUND

(2) In some engine architectures, aerodynamic flow distributors, such as scroll or volute structures, are used to receive combustion gases and to regulate them in a suitable manner before the combustion gases meet stator vanes or rotor blades of the downstream turbine(s). Such structures are subjected to thermal growth, which may have some various effects on surrounding components. Improvements are therefore sought.

SUMMARY

(3) In one aspect, there is provided an aircraft engine, comprising: a turbine including a turbine rotor rotatable about a central axis; a scroll case having an inlet fluidly connected to a source of combustion gases and an outlet fluidly connected to the turbine, and a conduit extending around the central axis from the inlet to the outlet; a bearing housing extending around the central axis; a turbine support case secured to the bearing housing, the scroll case disposed axially between the turbine support case and the bearing housing, the turbine support case having spokes distributed around the central axis and extending along a direction having an axial component relative to the central axis, the spokes extending through the scroll case and radially supported by the bearing housing; and heat shields distributed around the central axis in circumferential alignment with the spokes and extending axially through the scroll case for thermally shielding the spokes, a heat shield of the heat shields disposed around a spoke of the spokes, the heat shield defining an air gap surrounding the spoke inside the scroll case.

(4) The aircraft engine described above may include any of the following features, in any combinations.

(5) In some embodiments, the bearing housing includes a support flange, the heat shield secured to the support flange.

(6) In some embodiments, the spokes are secured to the bearing housing via first fasteners, the heat shields secured to the bearing housing via second fasteners being different than the first fasteners.

(7) In some embodiments, the bearing housing includes a support flange, the spokes and the heat shields secured to the support flange via the first fasteners and the second fasteners.

(8) In some embodiments, the heat shield extends from a first end to a second end, the first end secured to the bearing housing.

(9) In some embodiments, the second end abuts the spoke, the heat shield defining a ridge at or proximate the second end, the ridge extending through the air gap to abut the spoke.

(10) In some embodiments, the scroll case includes vanes extending in a direction having an axial component relative to the central axis and across the conduit.

(11) In some embodiments, each of the spokes extends within a respective one of the vanes.

(12) In some embodiments, second air gaps are between the spokes and the heat shields, the second air gaps surrounding the heat shields.

(13) In some embodiments, the spokes are free of connection to the vanes.

- (14) In another aspect, there is provided a turbine assembly, comprising: a turbine including a turbine rotor rotatable about a central axis; a support structure; a scroll case for receiving combustion gases and for directing the combustion gases to the turbine, the scroll case having a conduit extending around the central axis; and a turbine support case secured to the support structure, the turbine support case having spokes distributed around the central axis and extending along a direction having an axial component relative to the central axis, the spokes extending through the scroll case and radially supported by the support structure; and heat shields extending around the spokes, the spokes protected from heat of the combustion gases by both the heat shields and by air gaps extending around the spokes and defined between the spokes and the heat shields, the heat shields axially overlapping the scroll case.
- (15) The turbine assembly described above may include any of the following features, in any combinations.
- (16) In some embodiments, the heat shield extends from a first end to a second end, the first end secured to the support structure, the second end abutting the spoke.
- (17) In some embodiments, the heat shield defines a ridge at the second end, the ridge extending through the air gap to abut the spoke.
- (18) In some embodiments, the support structure includes a support flange, the heat shield secured to the support flange.
- (19) In some embodiments, the spokes are secured to the support structure via first fasteners, the heat shields secured to the support structure via second fasteners being different than the first fasteners.
- (20) In some embodiments, the support structure includes a support flange, the spokes and the heat shields secured to the support flange via the first fasteners and the second fasteners.
- (21) In some embodiments, the scroll case includes vanes extending in a direction having an axial component relative to the central axis and across the conduit.
- (22) In some embodiments, each of the spokes extends within a respective one of the vanes.
- (23) In some embodiments, second air gaps are between the spokes and the heat shields, the second air gaps surrounding the heat shields.
- (24) In some embodiments, the spokes are free of connection to the vanes.
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Description

DESCRIPTION OF THE DRAWINGS

- (1) Reference is now made to the accompanying figures in which:
- (2) FIG. 1 is a schematic side view of an aircraft engine;
- (3) FIG. 2 is a side cross-sectional view of a portion of the aircraft engine of FIG. 1 illustrating a hot section of the aircraft engine;
- (4) FIG. 3 is an enlarged view of a portion of FIG. 2;
- (5) FIG. 4 is a three dimensional exploded view of a turbine assembly for the aircraft engine of FIG. 1, including a bearing housing, a scroll case, and a turbine support case;
- (6) FIG. 5 is a cross-sectional view of the turbine support case of FIG. 4 taken on a plane normal to a central axis of the aircraft engine of FIG. 1;
- (7) FIG. 6 is an enlarged view of a portion of FIG. 3;
- (8) FIG. 7 is a cross-sectional view taken along line 7-7 on FIG. 6;
- (9) FIG. 8 is a three-dimensional cutaway view of a heat shield for the turbine support case;
- (10) FIG. 9 is another three-dimensional cutaway view of the heat shield; and
- (11) FIG. 10 is a three-dimensional partially transparent view of a portion of the turbine support case illustrating an assembly of the heat shield.

DETAILED DESCRIPTION

(12) Referring to FIG. 1, an aircraft engine **10** is schematically shown. The aircraft engine **10** comprises a thermal engine module **11** including one or more internal combustion engine(s), drivingly engaged to a rotatable load **12**, herein depicted as a propeller, via an output shaft **13**. It will be appreciated that the thermal engine module **11** may include any suitable engine, such as a gas turbine engine, a rotary engine, a piston engine, and so on. The output shaft **13** may correspond to an engine shaft of the thermal engine module **11**. The thermal engine module **11** may include any engine having at least one combustion chamber of varying volume. For instance, the thermal engine module **11** may comprise one or more piston engine(s) or one or more rotary engine(s) (e.g., Wankel engines). The aircraft engine **10** further includes a compressor **14** having a compressor inlet receiving ambient air from the environment E outside the aircraft engine **10** and a compressor outlet fluidly connected to an air inlet of the thermal engine module **11**. The compressor **14** outputs compressed air from the compressor outlet to the thermal engine module **11** via a compressed air conduit **16** and a manifold **17**. The compressed air conduit **16** and the manifold **17** may include any suitable arrangement of pipes configured to distribute compressed air between the different combustion chambers of the thermal engine module **11**. Any other suitable configurations used to supply compressed air to the thermal engine module **11** are contemplated without departing from the scope of the present disclosure. The aircraft engine **10** further includes a turbine assembly **15** having an axially facing turbine inlet **15A** fluidly connected to an engine outlet of the thermal engine module **11**. The turbine **15** has a turbine exhaust case **15B** via which combustion gases are expelled to the environment E. The turbine exhaust case **15B** may include a tailpipe or any other suitable structures (e.g., exhaust mixer) for discharging the combustion gases from the aircraft engine **10**. In some embodiments, the engine **10** may be a hybrid engine including an electric motor drivingly engaged to the output shaft **13** to assist the thermal engine module **11** in driving the output shaft **13** and the rotatable load (e.g., propeller **12**) mounted thereto.

(13) Referring jointly to FIGS. 1-2, in one or more embodiment(s), the turbine **15** includes an axial turbine having successive rows of rotor(s) **15C** and stator(s) **15D** disposed in alternation along a central axis A of the aircraft engine **10**. The rotor(s) **15C** may include rotor blades mounted to rotor discs. The stator(s) **15D** may include stator vanes secured at opposite ends to inner and outer shrouds. In other words, the turbine **15** may include a plurality of stages each including a stator and a rotor. The rotors **15C** of the turbine **15** are in driving engagement with a turbine shaft **15E**. The turbine shaft **15E** may be drivingly engaged to the output shaft **13**, which may correspond to the engine shaft of the thermal engine module **11**. Therefore, the turbine **15** may compound power with the thermal engine module **11** to drive the rotatable load **12**. In other words, the turbine shaft **15E** may be drivingly engaged to the engine shaft of the thermal engine module **11** via suitable gearing. In the embodiment shown, the turbine shaft **15E** is drivingly engaged to a compressor shaft of the compressor **14**. Thus, the turbine **15** may drive both the rotatable load **12** and the compressor **14**. In the exemplified embodiment, the engine shaft of the thermal engine module **11**, the output shaft **13**, and the turbine shaft **15E** are all coaxial about the central axis A. However, in other configurations, the turbine **15** and/or the compressor **14** may have respective shafts radially offset from one another relative to the central axis A.

(14) As shown in FIG. 1, the engine outlet of the thermal engine module **11** is fluidly connected to an exhaust manifold **18** that receives combustion gases outputted by the combustion chambers or by a combustor of the thermal engine module **11**. The exhaust manifold **18** collects the combustion gases from the different combustion chambers and flows these combustion gases to a combustion engine exhaust pipe **19** that feeds the combustion gases to the turbine **15**. In other words, the engine outlet of the thermal engine module **11** is fluidly connected to the turbine inlet **15A** via the exhaust manifold **18** and the combustion engine exhaust pipe **19**. Any other suitable configurations used to supply combustion gases to the turbine **15** are contemplated without departing from the scope of the present disclosure.

(15) As schematically depicted by the flow arrows in FIG. 1, the combustion gases are flowing

within the combustion engine exhaust pipe **19** and reach the turbine **15** in a direction being mainly radial relative to the central axis A and which may include a circumferential component relative to the central axis A. However, the turbine **15** includes an axial turbine and therefore the turbine inlet **15A** receives the combustion gases along a direction being mainly axial relative to the central axis A. To redirect the combustion gases from a direction being mainly radial to a direction being mainly axial, that is, to decrease a radial component of a direction of the combustion gases, the aircraft engine **10** further includes a scroll case **20** that regulates and reorients the combustion gases so that they meet an upstream most of the stages of the turbine **15** at the most appropriate angle of attack. In the embodiment shown, the flow of combustion gases exiting the scroll case **20** meets a first stage rotor **15C** of the turbine **15** before meeting a stator thereof. The scroll case **20** may therefore be used to adequately orient the combustion gases at the most appropriate angle to meet the upstream-most airfoils of the turbine **15**, which are herein part of one of the first stage rotors **15C**.

(16) Referring to FIG. 3, as shown in the exemplified embodiment, the scroll case **20** may be provided in form of a unitary body or mono-case comprising a conduit **21** extending around the central axis A from an inlet **22** to an outlet **23**. The inlet **22** is fluidly connected to the combustion engine exhaust pipe **19**, whereas the outlet **23** is fluidly connected to the turbine inlet **15A** (FIG. 2) of the turbine **15**. According to the illustrated embodiment, the inlet **22** of the conduit **21** has a tangential component and the outlet **23** is an annular outlet facing axially in a rearward direction and in alignment with an annular gas path **15F** of the turbine **15**. This configuration allows injecting the combustion gases in a direction being mainly axial relative to the central axis A to meet the axial inlet of the turbine **15**. Vanes **24** may be provided in the conduit **21** to direct and regulate the flow of combustion gases. The vanes **24** may be omitted in some embodiments. The conduit **21** of the scroll case **20** is in this embodiment disposed axially forwardly of the turbine **15**.

(17) The conduit **21** comprises a non-axisymmetric portion extending downstream from the inlet **22** and spiraling towards the central axis A. As it progresses circumferentially around the central axis A, the non-axisymmetric portion of the conduit **21** transitions or merges with an axisymmetric portion, which forms a 360 degrees axisymmetric structure around the central axis A. The axisymmetric portion extends downstream from the non-axisymmetric portion to the outlet **23**.

(18) The inventors have found that in engine running conditions, the thermal distortions are non-uniform in the non-axisymmetric portion of the scroll case **20**. Consequently, using the scroll case **20** to secure the turbine exhaust case **15B** may increase tip clearance of the rotors **15C** of the turbine **15**. In other words, radial thermal growth of the scroll case **20** during use of the engine may move the turbine exhaust case **15B** radially outwardly, thus pulling radially on shrouds disposed around the rotors **15C**. This may increase tip clearance and, as a result, may impair performance. As will be seen hereafter, a turbine support case arrangement may be used to alleviate these drawbacks.

(19) As illustrated on FIG. 3, a compressor case **14A** of the compressor **14** is radially supported by a bearing housing **30**. It will be appreciated that that any suitable support structure may be used for supporting the compressor case **14A**. For instance, the support structure may be any static component of the engine, such as a support flange and so on. Bearings **31** are rollingly engaged to the bearing housing **30** and radially support a shaft of the engine. The scroll case **20** is secured to a rear end **32** of the bearing housing **30**. In the exemplified embodiment, the scroll case **20** has a radially-inner wall **25** that defines a flange at its rear end. The flange of the radially-inner wall **25** is received within an annular groove defined by the rear end **32** of the bearing housing **30**. Other configurations are however contemplated. Therefore, the scroll case **20** may not rely on the turbine exhaust case **15B** for structural support.

(20) In the disclosed embodiment, a turbine support case **40** is used to secure the turbine exhaust case **15B** to the compressor case **14A** of the compressor **14**. As will be explained below, the turbine support case **40** is independent from the scroll case **20** such that thermal growth of the scroll case

20 may not be transmitted to the turbine exhaust case **15B**. Therefore, the turbine exhaust case **15B** is secured to the compressor case **14A** via the turbine support case **40** independently of the scroll case **20**. In the present disclosure, the expression “independent” or “independently” in “independently of the scroll case **20**” implies that a load path extends from the compressor case **14A** to the turbine exhaust case **15B** through the turbine support case **40** without intersecting the scroll case **20**. The scroll case **20** is therefore free from intersection to the load path from the compressor case **14A** to the turbine exhaust case **15B**. The scroll case **20** is thus not part of the load path from the compressor case **14A** to the turbine exhaust case **15B** and loads generated by the turbine **15** on the turbine exhaust case **15B** are transmitted to the compressor case **14B** via the turbine support case **40** without assistance from the scroll case **20**. The scroll case **20** is thus outside the load path that extends through the turbine support case **40**. The scroll case **20** may thus be structurally floating relative to the turbine support case **40**.

(21) Referring to FIG. **4**, the turbine support case **40** has a portion that axially overlaps the scroll case **20** and is secured to an annular member **41**, which is itself secured to the bearing housing **30** or any other suitable support structure. More specifically, the annular member **41** has a flange **42** secured (e.g., bolted) to a first flange **33** of the bearing housing **30**. The bearing housing **30** further has a second flange **34**, which may be disposed radially outwardly of the first flange **33** and axially offset from the first flange **33**, for being secured (e.g., bolted) to a mating flange of the compressor case **14A**.

(22) The turbine support case **40** includes a wall **43** extending around the central axis A. The wall **43** may be cylindrical, frustoconical, or any other suitable shape. The wall **43** may extend a full circumference around the central axis A. The turbine support case **40** further includes spokes **44** protruding from the wall **43**. More specifically, the turbine support case **40** includes an annular axial wall **45** extending radially inwardly from the wall **43**. The spokes **44** protrude in a direction having an axial component relative to the central axis A from the annular axial wall **45** and away from the wall **43**. The spokes **44** may be parallel to the central axis A. An annular flange **46** is provided at a rear end of the wall **43** and is secured (e.g., bolted) to a mating flange **15G** (FIG. **3**) of the turbine exhaust case **15B**.

(23) As shown in FIG. **3**, the wall **43** axially overlaps at least a portion of the turbine **15**. A containment ring **50** may be secured to the flange **15G** of the turbine exhaust case **15B** via containment ring flange **51**, which may be sandwiched between the annular flange **46** of the turbine support case **40** and the flange **15G** of the turbine exhaust case **15B**. The containment ring **50** is, in this embodiment, disposed radially between the wall **43** of the turbine support case **40** and at least one of the rotors **15C** of the turbine **15**.

(24) The spokes **44**, six in the illustrated embodiment, but more or less may be used, extend from proximal ends **44A** at the annular axial wall **45** to distal ends **44B**. The distal ends **44B** of the spokes **44** are secured to the annular member **41** as will be explained further below. The distal ends **44B** of the spokes define threaded apertures **44C** (FIG. **5**) threadingly engageable by fasteners **47** (e.g., bolts) extending through correspondingly-shaped apertures defined through the annular member **41** and threadingly engaged to the threaded apertures **44C** for securing the spokes **44** to the annular member **41**, which is itself secured to the bearing housing **30**.

(25) Referring to FIGS. **4** and **5**, in the embodiment shown, each of the spokes **44** is received within a respective one of the hollow vanes **24** of the scroll case **20**. The spokes **44** therefore axially overlap the vanes **24**. Thus, the spokes **44** may be isolated from combustion gases flowing through the scroll case **20** by the vanes **24**. The spokes **44** may be free of connection to the vanes **24**. In other words, outer surfaces of the spokes **44** may be free of contact with inner surfaces of the vanes **24**. An annular gap may be provided between the inner surface of each vanes **24** and the associated spokes **44** extending internally therethrough. The vanes **24** may move axially, radially, and/or circumferentially relative to the spokes **44** without transferring any forces to the spokes **44**, and vice versa. Put differently, the scroll case **20** is free from direct connection to the turbine

support case **40**. In other words, the scroll case **20** is free of contact, attachment, so on with the turbine support case **40**. The spokes **44** of this embodiment have an elongated, airfoil-like shape to substantially match a shape of the vanes **24**. However, the shape of the spokes **44** may be different. The spokes **44** may be circular, oval, square, rectangular in cross-section and so on, without departing from the scope of the present disclosure.

(26) In some conditions, a temperature of the combustion gases flowing in the scroll case **20** may be very high and it may be required to thermally shield the spokes **44** from the combustion gases. The thermal conductivity and the heat transfer may affect the spokes. Moreover, the free space between the vanes **24** and the spokes **44** is limited. The turbine support case **40** of the present disclosure may at least partially alleviate these drawbacks.

(27) Referring to FIGS. 5-7, in the embodiment shown, heat shields **60** are provided to protect the spokes **44** from the hot combustion gases. As shown in FIG. 6, the heat shields **60** axially overlap the scroll case **20** relative to the central axis A of the engine. The heat shields **60** extend around the spokes **44**. More specifically, each of the heat shields **60** extends around a respective one of the spokes **44**. As depicted in FIG. 7, first air gaps G1 are thus created between the spokes **44** and the heat shields **60**. Also, second air gaps G2 are created between the heat shields **60** and the vanes **24**. Consequently, a heat transfer path from the combustion gases to the spokes **44** extends through a thickness of a material of the vanes **24**, through the second air gap G2, through a thickness of a material of the heat shields **60**, which may be Hastalloy™, Inconel™, or any suitable material, and through the first air gap G1. The heat shield **60** may therefore impede at least partially heat transfer that would otherwise occur by convection. Also, the heat shield **60** may at least partially impede a heat transfer via radiation from the vanes **24** to the spokes **44**. In other words, the spokes **44** may be protected from heat of the combustion gases by both of the heat shields **60** and by the first air gaps G1 extending around the spokes **44** and defined between the spokes **44** and the heat shields **60**.

(28) One of the heat shields **60** is described below using the singular form, but it will be appreciated that this description may apply to all of the heat shields **60**. The heat shield **60** extends from a first end **61** to a second end **62**. The first end **61** is secured to the bearing housing **30**. More particularly, in the embodiment shown, the first end **61** is secured to the bearing housing **30** via the annular member **41**. The second end **62** is located proximate the annular axial wall **45**. In this embodiment, the second end **62** is in abutment with one of the spokes **44**. More specifically, the heat shield defines a ridge **63** at or proximate the second end **62**. The ridge **63** extends through the first air gap G1 to abut the spoke **44**. This ridge **63** need not be fully annular and may contact the spoke **44** only at a few spaced-apart locations. The ridge **63** may alternatively be annular and define a contact line with the spoke **44**, the contact line extending all around the spoke **44**. The abutment between the second end **62** of the heat shield **60** and the spoke **44** may prevent undesired vibrations of the heat shield **60**. In some embodiments, the second end **62** of the heat shield **60** may be free of contact with the spoke **44**.

(29) Referring to FIGS. 8-10, the heat shield **60** includes a flange **64** at the first end **61**. The flange **64** defines one or more apertures **64A** for receiving fasteners to secure the heat shield **60** to the annular member **41** or any other support flange. As shown in FIG. 10, the flange **64** is secured to the annular member **41** via a second set of fasteners **48**, which are different than the fasteners **47** (FIG. 4) used to secure the distal ends **44B** of the spokes **44** to the annular member **41**. In other words, the fasteners **47** used to secure the spokes **44** to the annular member **41** are different than the fasteners **48** used to secure the heat shields **60** to the annular member **41**. This may facilitate an assembly procedure. Indeed, the heat shield **60** may be secured first to the annular member **41** using the fasteners **48**. At which point, the scroll case **20** may be inserted such that each of the vanes **24** receives a respective one of the heat shields **60** therein. Then, the turbine support case **40** may be inserted by inserting each of the spokes **44** within a respective one of the heat shields **60** and securing, with the fasteners **47**, the distal ends **44B** of the spokes **44** to the annular member **41**.

(30) It is noted that various connections are set forth between elements in the preceding description

and in the drawings. It is noted that these connections are general and, unless specified otherwise, may be direct or indirect and that this specification is not intended to be limiting in this respect. A coupling between two or more entities may refer to a direct connection or an indirect connection. An indirect connection may incorporate one or more intervening entities. The term “connected” or “coupled to” may therefore include both direct coupling (in which two elements that are coupled to each other contact each other) and indirect coupling (in which at least one additional element is located between the two elements).

(31) It is further noted that various method or process steps for embodiments of the present disclosure are described in the preceding description and drawings. The description may present the method and/or process steps as a particular sequence. However, to the extent that the method or process does not rely on the particular order of steps set forth herein, the method or process should not be limited to the particular sequence of steps described. As one of ordinary skill in the art would appreciate, other sequences of steps may be possible. Therefore, the particular order of the steps set forth in the description should not be construed as a limitation.

(32) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. As used herein, the terms “comprises”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(33) While various aspects of the present disclosure have been disclosed, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the present disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these particular features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the present disclosure. References to “various embodiments,” “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. The use of the indefinite article “a” as used herein with reference to a particular element is intended to encompass “one or more” such elements, and similarly the use of the definite article “the” in reference to a particular element is not intended to exclude the possibility that multiple of such elements may be present.

(34) The embodiments described in this document provide non-limiting examples of possible implementations of the present technology. Upon review of the present disclosure, a person of ordinary skill in the art will recognize that changes may be made to the embodiments described herein without departing from the scope of the present technology. Yet further modifications could be implemented by a person of ordinary skill in the art in view of the present disclosure, which modifications would be within the scope of the present technology.

Claims

1. An aircraft engine, comprising: a turbine including a turbine rotor rotatable about a central axis; a scroll case having an inlet fluidly connected to a source of combustion gases and an outlet fluidly connected to the turbine, and a conduit extending around the central axis from the inlet to the outlet; a bearing housing extending around the central axis; a turbine support case secured to the bearing housing, the scroll case disposed axially between the turbine support case and the bearing housing, the turbine support case having spokes distributed around the central axis and extending

along a direction having an axial component relative to the central axis, each of the spokes extending through the scroll case and radially supported by the bearing housing; and heat shields distributed around the central axis in circumferential alignment with the spokes and extending axially through the scroll case for thermally shielding the spokes, a heat shield of the heat shields disposed around a spoke of the spokes, the heat shield defining an air gap surrounding the spoke inside the scroll case.

2. The aircraft engine of claim 1, wherein the bearing housing includes a support flange, the heat shield secured to the support flange.
3. The aircraft engine of claim 2, wherein the spokes are secured to the bearing housing via first fasteners, the heat shields secured to the bearing housing via second fasteners being different than the first fasteners.
4. The aircraft engine of claim 3, wherein the bearing housing includes a support flange, the spokes and the heat shields secured to the support flange via the first fasteners and the second fasteners.
5. The aircraft engine of claim 1, wherein the heat shield extends from a first end to a second end, the first end secured to the bearing housing.
6. The aircraft engine of claim 5, wherein the second end abuts the spoke, the heat shield defining a ridge at or proximate the second end, the ridge extending through the air gap to abut the spoke.
7. The aircraft engine of claim 1, wherein the scroll case includes vanes extending in a direction having an axial component relative to the central axis and across the conduit.
8. The aircraft engine of claim 7, wherein each of the spokes extends within a respective one of the vanes.
9. The aircraft engine of claim 8, comprising second air gaps between each of the spokes and the respective one of the heat shields, the second air gaps surrounding the heat shields.
10. The aircraft engine of claim 8, wherein each of the spokes are free of connection to the respective one of the vanes.
11. A turbine assembly, comprising: a turbine including a turbine rotor rotatable about a central axis; a support structure; a scroll case for receiving combustion gases and for directing the combustion gases to the turbine, the scroll case having a conduit extending around the central axis; and a turbine support case secured to the support structure, the turbine support case having spokes distributed around the central axis and extending along a direction having an axial component relative to the central axis, each of the spokes extending through the scroll case and radially supported by the support structure; and heat shields extending around each of the spokes, the spokes protected from heat of the combustion gases by both the heat shields and by air gaps extending around the spokes and defined between each of the spokes and the respective one of the heat shields, the heat shields axially overlapping the scroll case.
12. The turbine assembly of claim 11, wherein a heat shield of each of the heat shields extends from a first end to a second end, the first end secured to the support structure, the second end abutting a spoke of the spokes.
13. The turbine assembly of claim 12, wherein the heat shield defines a ridge at the second end, the ridge extending through the air gap to abut the spoke.
14. The turbine assembly of claim 11, wherein the support structure includes a support flange, the heat shields secured to the support flange.
15. The turbine assembly of claim 14, wherein the spokes are secured to the support structure via first fasteners, the heat shields secured to the support structure via second fasteners being different than the first fasteners.
16. The turbine assembly of claim 15, wherein the support structure includes a support flange, the spokes and the heat shields secured to the support flange via the first fasteners and the second fasteners.
17. The turbine assembly of claim 11, wherein the scroll case includes vanes extending in a direction having an axial component relative to the central axis and across the conduit.

18. The turbine assembly of claim 17, wherein each of the spokes extends within a respective one of the vanes.
 19. The turbine assembly of claim 18, comprising second air gaps between each of the vanes and the respective one of the heat shields, the second air gaps surrounding the heat shields.
 20. The turbine assembly of claim 18, wherein each of the spokes are free of connection to the respective one of the vanes.
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